

Harsh Garg

Punjab, India | +91 9041594669 | gargharsh2307@gmail.com
GitHub: github.com/Harshgg1 | LinkedIn: linkedin.com/in/harsh-garg-71a465328/

TECHNICAL SKILLS

Languages: C++, JavaScript, TypeScript, C

Frontend: React.js, Next.js, HTML, CSS, TailwindCSS

Backend: Node.js, Express.js, REST APIs, Authentication (JWT), WebSockets

Databases: MongoDB, PostgreSQL, Redis

DevOps: Docker, Turborepo

Core CS: Data Structures & Algorithms, OOP, DBMS, Operating Systems, Computer Networks

Tools: Git, GitHub, VS Code

PROJECTS

CourseMint - Full Stack Course Management Platform

- Built a full-stack course management platform enabling users to browse, enroll in, and manage courses through a dynamic React interface.
- Designed and implemented RESTful APIs using Node.js and Express for course CRUD operations and backend business logic.
- Integrated MongoDB for persistent storage of course and user-related data.
- Developed responsive UI components and reusable frontend architecture using React.
- Structured backend modules and routing to improve scalability and maintainability.

Real-Time Chat System with Redis

- Developed a real-time messaging backend supporting concurrent communication between multiple users.
- Implemented WebSocket-based bidirectional communication for low-latency message delivery.
- Integrated Redis Pub/Sub for efficient message broadcasting across active connections.
- Designed scalable connection and message-handling architecture.
- Built the foundation for horizontally scalable real-time communication systems.

ACHIEVEMENTS

- Demonstrated strong proficiency in Data Structures and Algorithms by consistently solving a wide variety of complex problems across major competitive coding platforms.
- Strong foundation in problem solving, algorithm design, and complexity analysis.
- Actively learning advanced backend engineering, distributed systems, and scalable application design.

EDUCATION

Thapar Institute of Engineering & Technology

Bachelor of Engineering in Computer Engineering

2024-2028

Relevant Coursework: Data Structures & Algorithms, Database Systems, Operating Systems, Computer Networks